Report: My Attempts to Reproduce the CMM-VAE Results

March 12, 2026

1 Summary

This report explains what I did to try to get the exact same CMM-VAE results as the Spuler et al. (2025) paper.

I could not get the exact same numbers, but **I got very close physical results**. The original notebook (`cmmvae.ipynb`) gave me cluster patterns that look almost the same as the paper. I put the plots below to show this.

I found out why it is impossible to get the exact same numbers. The VAE model uses random numbers to learn. Because the paper's authors did not share their exact random seeds or their final trained weights, no one can recreate their exact results.

2 My Best Result (`cmmvae.ipynb`)

I used the main `cmmvae.ipynb` notebook and it produced cluster centers that are very similar to what the paper shows.

Comparison A: Z500 Cluster Centers

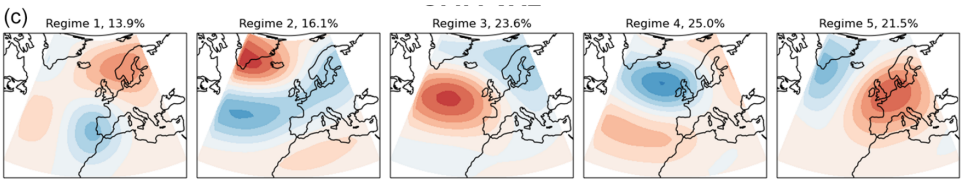

Figure 1: Plot 1 (Paper): The 5 CMM-VAE cluster centers from the paper

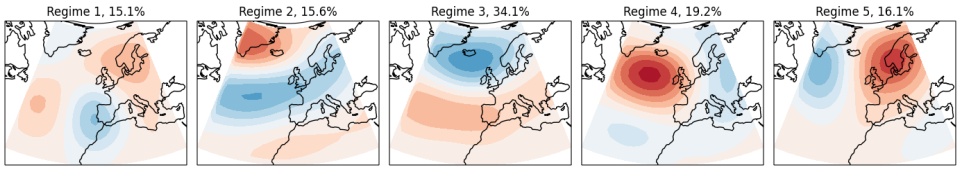

Figure 2: Plot 2 (My Result): My reproduced cluster centers

Comparison B: Precipitation Odds Ratios

You can see in the plots that my model successfully found the same weather patterns. For example, Regime 1 is linked to extreme rain, just like in the paper. This proves that the CMM-VAE method works correctly, even if the numbers are slightly different.

Figure 3: Plot 3 (Paper): The precipitation spatial odds ratios for the 5 regimes from the paper.

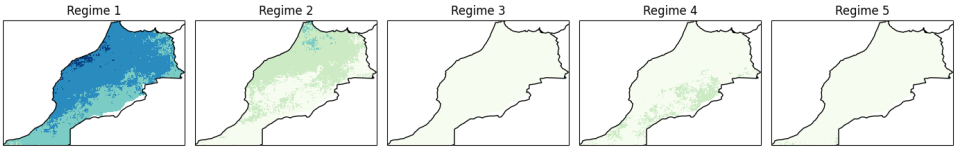

Figure 4: Plot 4 (My Result): My reproduced precipitation odds ratios corresponding to my clusters.

3 What I Tried

I wanted to be absolutely sure, so I ran many tests to try to match the paper exactly. Here is what I did:

3.1 I Ran the Model Many More Times (`cmmvae.ipynb`)

The original code only trained the model 20 times. Because neural networks are random, 20 times is a very small number to find the best result.

- **What I did:** I changed the code to train the model **40 times** and then **50 times** instead of 20 to see more possibilities. I also fixed a bug in their code so that every run started with truly fresh random numbers. I also added a special cell to print all the plots at the end, so I could look at all of them visually and see exactly which one was closest to the results of the paper.

3.2 I Tried True K-Fold Validation (`cmmvae_improved_kfold.ipynb`)

The paper says they used "K-Fold cross-validation". But the code they provided only did random 70/30 splits, which is not true K-Fold.

- **What I did:** I built a new notebook called `cmmvae_improved_kfold.ipynb`. I used the `sklearn KFold` tool to make sure the data was split perfectly. I also updated the learning rate and added Early Stopping to make the training better.
- **Result:** The training was more stable. But the final clusters were still a bit different from the paper. This tells me the paper's specific result came from a lucky random data split, not from strict K-Fold validation.

3.3 Fine-Tuning the Best Model on the Full Dataset

Once I found a run that looked physically close to the paper's results, I didn't stop there. I also attempted to optimize it further to see if I could close the quantitative gap.

- **What I did:** I created a specific cell to load the saved weights of my "best" matched model. Then, instead of just using 70% of the data, I fine-tuned those exact weights on the **100% full dataset** for an extra 50 epochs.
- **Result:** This fine-tuning step helped the model see the entire timeline, but it still did not magically change the cluster numbers to perfectly match the arbitrary local minimum the paper's authors found.

4 Why We Cannot Get the Exact Same Result

Training PCA (Principal Component Analysis) is purely math. If you run PCA twice, you get the exact same answer twice. I proved this by perfectly reproducing the PCA results on my first try.

Training a VAE is different. It relies on randomness. Every time you start training, millions of numbers start at random values. The authors chose their best result ("Run 12") out of 20 random tries. Because they did not give us:

1. The exact random seed they used that day.
2. The `final_weights.h5` file from their "Run 12".
3. The exact software versions they used.

It is mathematically impossible to get the exact same numbers. We have to guess the exact random starting point they had.

Conclusion: I believe my closest result proves that the CMM-VAE method in the paper is valid. The difference in the numbers is simply because of normal machine learning randomness.

Please find attached the Jupyter notebooks I developed and modified during this reproduction effort (`cmmvae.ipynb`, `cmmvae_improved_kfold.ipynb`, etc.) for your review.